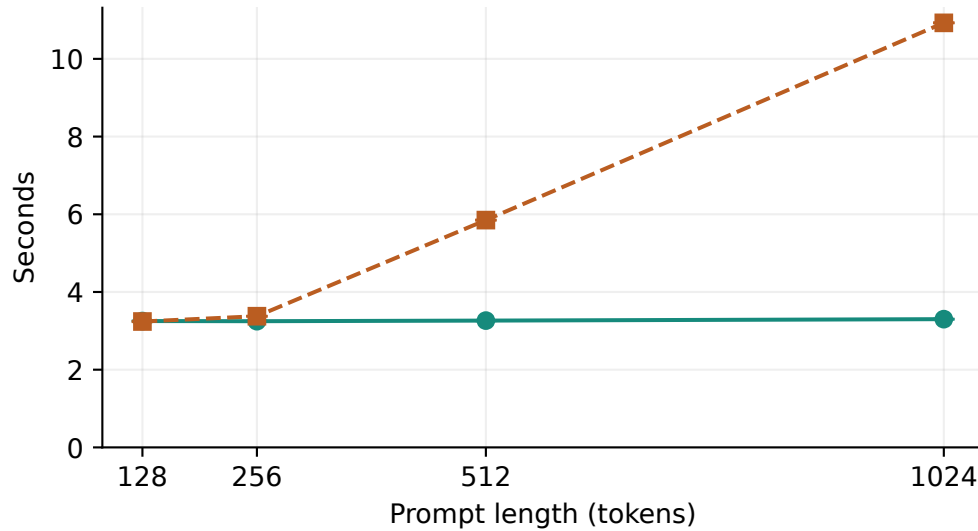


KV Cache ON/OFF: prompt length sweep

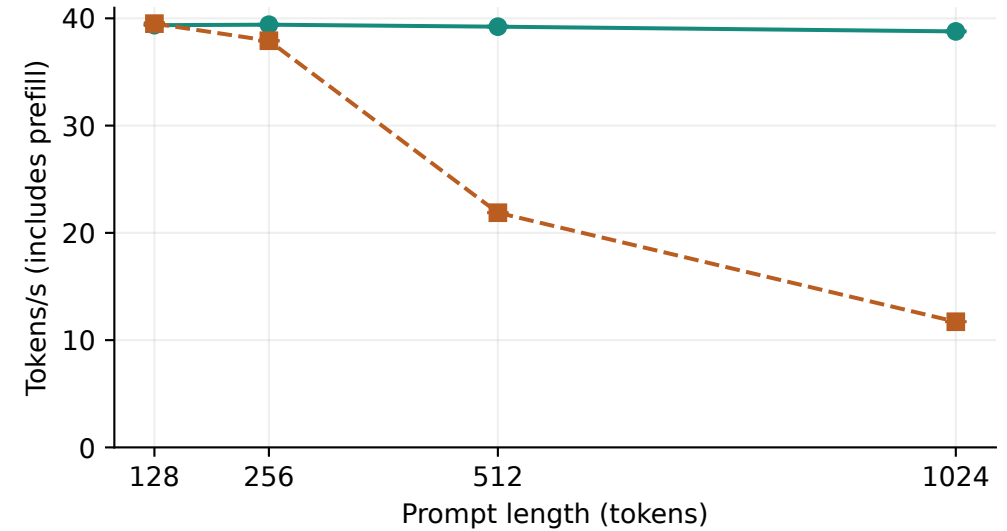
Qwen2.5-1.5B | RTX 3090 | BF16 | eager | generation 128 | batch 1

Cache ON Cache OFF

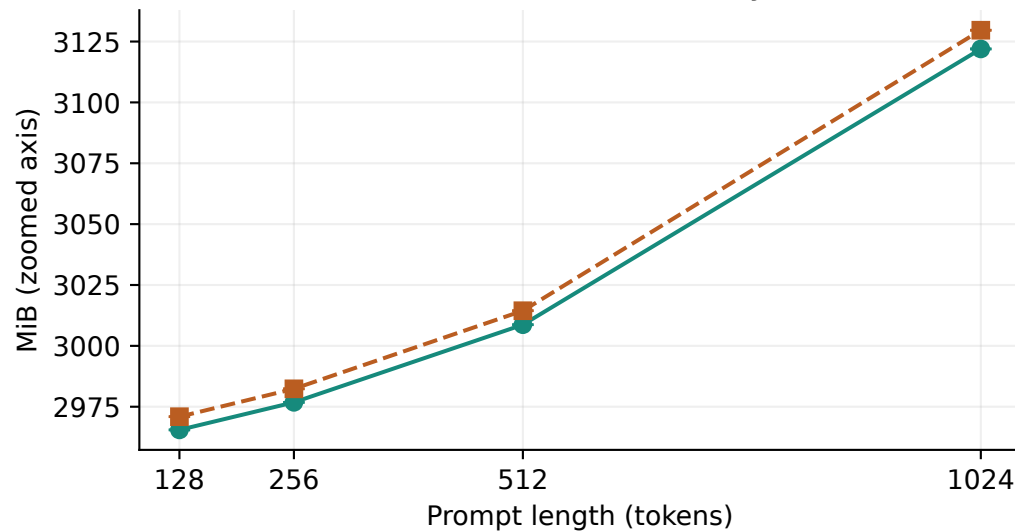
Total generation latency



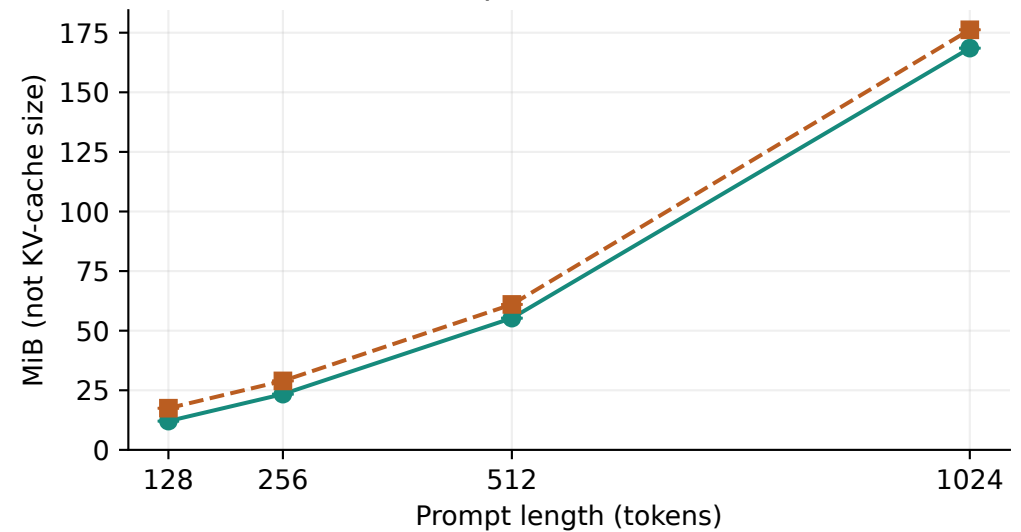
Generated-token throughput



Peak allocated memory



Allocated peak above baseline



Means $\pm$ sample SD (n=5, ddof=1). Warm-up excluded. Inputs repeat the original 128 token IDs.

Latency speedup (OFF / ON) by prompt length: 128: 1.00x; 256: 1.04x; 512: 1.79x; 1024: 3.31x

Outputs match across all ten measured runs at each length. Peak memory includes temporary tensors.